Complete Guide to Dockerfile Parameters 🐳🚀

A Dockerfile is a script containing a set of instructions to build a Docker image. It automates the creation of lightweight, portable, and consistent containerized applications. Below is a detailed breakdown of all important Dockerfile parameters, their syntax, and examples.

1️⃣ FROM (Base Image)
Defines the base image for the container.
📌 Every Dockerfile must start with FROM.
✅ Syntax: FROM <image>:<tag>
✅ Example: FROM ubuntu:20.04
(Uses Ubuntu 20.04 as the base image.)

2️⃣ LABEL (Metadata)
Adds metadata (author, description, version, etc.) to an image.
✅ Syntax: LABEL key="value"
✅ Example: LABEL maintainer="John Doe"
LABEL description="This is a sample web application"

3️⃣ WORKDIR (Set Working Directory)
Sets the working directory for subsequent commands.
✅ Syntax: WORKDIR /app
✅ Example: WORKDIR /usr/src/app
(All commands like RUN, CMD, COPY, etc., will now run inside /usr/src/app.)

4️⃣ COPY (Copy Files from Host to Container)
Copies files from the local system to the container image.
✅ Syntax: COPY <source> <destination>
✅ Example: COPY index.html /usr/share/nginx/html/
(Copies index.html from the host to the Nginx container's web directory.)

5️⃣ ADD (Similar to COPY, but More Powerful)
Like COPY, but also extracts compressed files (.tar, .zip, etc.) and supports remote URLs.
✅ Syntax: ADD <source> <destination>
✅ Example: ADD app.tar.gz /usr/src/app/
(Extracts app.tar.gz into /usr/src/app/ inside the container.)

6️⃣ RUN (Execute Commands in the Image Build Process)
Runs a shell command while building the image.
✅ Syntax: RUN <command>
✅ Example: RUN apt-get update && apt-get install -y nginx
(Updates package lists and installs Nginx in the image.)

7️⃣ CMD (Define Default Container Startup Command)
Specifies the default command to run inside the container at runtime.
✅ Syntax: CMD ["executable", "param1", "param2"]
✅ Example: CMD ["nginx", "-g", "daemon off;"]
(Starts Nginx when the container runs.)
🛑 Difference between CMD and RUN
RUN executes during build time (inside the image).
CMD executes when the container starts (at runtime).

8️⃣ ENTRYPOINT (Define a Fixed Executable for the Container)
Similar to CMD, but it cannot be overridden when running a container.
✅ Syntax: ENTRYPOINT ["executable", "param1", "param2"]
✅ Example: ENTRYPOINT ["/bin/bash", "-c"]
(Forces the container to run Bash shell when started.)

🛑 Difference between CMD and ENTRYPOINT
CMD can be overridden when running the container.
ENTRYPOINT always executes, even if arguments are provided.

9️⃣ EXPOSE (Define Container Listening Port)
Declares which port the application will use inside the container.
✅ Syntax: EXPOSE <port>
✅ Example: EXPOSE 80
(Tells Docker that the application inside uses port 80.)
🚀 To publish a port while running the container: docker run -p 8080:80 my-image
(Maps port 80 in the container to port 8080 on the host.)

🔟 ENV (Set Environment Variables)
Defines environment variables inside the container.
✅ Syntax: ENV <key>=<value>
✅ Example: ENV APP_ENV=production
(Sets APP_ENV to production inside the container.)

1️⃣1️⃣ VOLUME (Persistent Storage for Containers)
Creates a mount point for persistent storage.
✅ Syntax: VOLUME ["/data"]
✅ Example: VOLUME ["/var/log/nginx"]
(Creates a persistent volume for Nginx logs.)

1️⃣2️⃣ ARG (Define Build-Time Variables)
Defines a variable that is used only during build time.
✅ Syntax: ARG <variable_name>
✅ Example: ARG APP_VERSION=1.0
RUN echo "App version: $APP_VERSION"
🚀 To pass an argument while building an image: docker build --build-arg APP_VERSION=2.0 -t my-app .

1️⃣3️⃣ HEALTHCHECK (Monitor Container Health)
Defines a health check to determine if a container is working properly.
✅ Syntax: HEALTHCHECK --interval=<time> --timeout=<time> CMD <command>
✅ Example: HEALTHCHECK --interval=30s --timeout=5s --retries=3 CMD curl -f http://localhost || exit 1
(Checks every 30s if the web server is responding.)

1️⃣4️⃣ USER (Run Commands as a Specific User)
Sets the user to execute commands inside the container.
✅ Syntax: USER <username or UID>
✅ Example: USER nginx
(Runs the container processes as the nginx user.)

1️⃣5️⃣ ONBUILD (Trigger Actions for Child Images)
Defines a command that executes only when a child image is built from this image.
✅ Syntax: ONBUILD <command>
✅ Example: ONBUILD COPY . /app